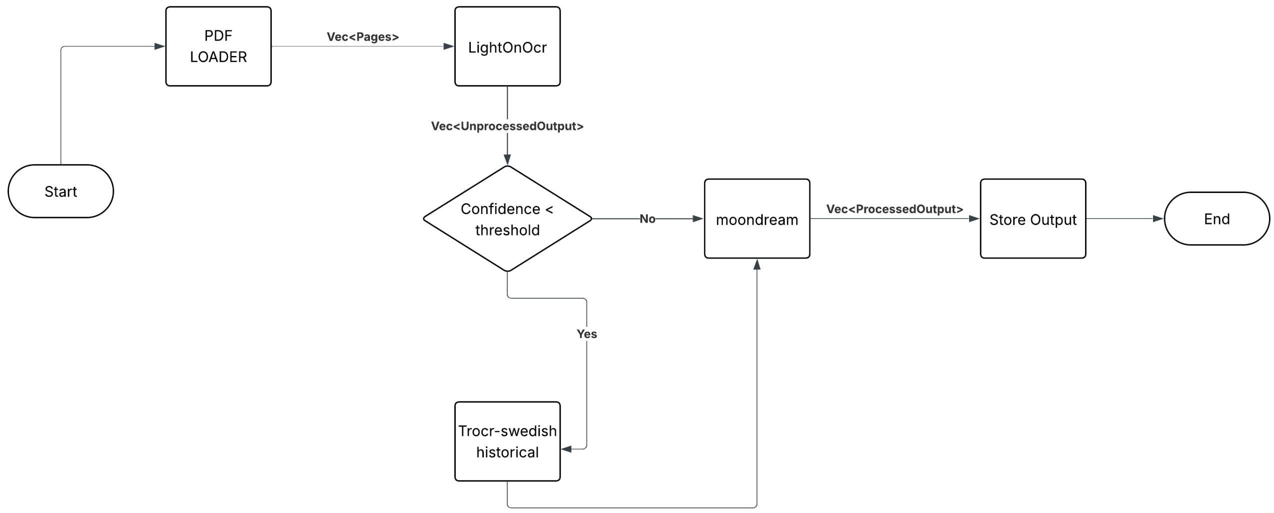
1. Overview

This report describes the multi-model OCR pipeline developed during the Softwerk hackathon for transcribing Swedish historical documents. The system was implemented entirely in Rust using the Candle machine learning framework, targeting GPU-accelerated inference tested a laptop NVIDIA RTX 3070 Ti.

Pipeline:



2.1 High-Level Flow

- PDF and image files are ingested from a data directory, PDFs converted to PNG via pdftoppm at 200 DPI.
- All pages are processed by LightOnOCR (full-page, one-shot inference)
- If LightOnOcr is not confident with the transcription the page must be handwritten. ~~Send page to handwritten pipeline.~~
- ~~Handwritten pages are segmented into lines and processed by TrOCR line-by-line~~ (Due to a bug found right before submission causing all files to be treated we have decided to remove this step.)
- Image regions detected in LightOnOCR output are cropped and described by Moondream1
- Final output is written to per-page text files in data/output/

2.2 Model Stack

Models	Description
LightOnOCR[1]	lightonai/LightOnOCR-2-1B-bbox-soup 1B parameter vision-language model based on Qwen3 + SigLIP. Handles full-page printed document transcription and bounding box detection for embedded images.

Models	Description
TrOCR[2]	Riksarkivet/trocr-base-handwritten-hist-swe-2 ViT encoder + TrOCR decoder fine-tuned on Swedish historical handwriting. Operates line-by-line after projection profile segmentation.
Moondream1[3]	vikhyatk/moondream1 1.6B parameter VLM (ViT + Phi). Used to describe cropped image regions detected by LightOnOCR, with forensic context prompt derived from surrounding document text.

2.3 Memory Management

All three models cannot fit in 8GB VRAM simultaneously. The pipeline uses Rust scoped blocks to ensure models are dropped from memory before the next model is loaded:

- LightOnOCR loads, processes all pages, then is dropped
- Moondream1 loads only after LightOnOCR is confirmed freed
- ~~TrOCR is loaded only when handwritten pages are detected~~

3. Implementation Details

3.1 LightOnOCR Integration

LightOnOCR required a complete ground-up implementation in Rust as no Candle port existed. Key components built from scratch:

- Integrating built in pixtral implementation from candle as Vision encoder.
- Spatial merge projector: 2×2 patch merging reducing 4840 vision tokens to 2145 IMAGE_PAD tokens
- Qwen3-based causal language model with grouped query attention and KV cache
- Custom token sequence builder matching the model's Jinja chat template exactly

~~3.2 TrOCR Integration~~

~~TrOCR uses a ViT encoder (768 hidden dimensions, 12 layers, 384×384 input) paired with a TrOCR decoder (1024 d_model, 12 layers).Line segmentation was uses horizontal projection profiling summing dark pixels per row to detect text bands and gaps sufficient for clean printed documents.~~

3.3 Moondream1 Integration

Moondream1 was selected over Moondream2 due to architecture compatibility with Candle's existing implementation. Moondream2 uses a fundamentally different architecture including tiled image crops, GQA attention, and a custom text transformer incompatible with the Phi-based Candle port.

Image regions are described using a forensic investigation prompt that incorporates surrounding document text as context, improving description relevance for historical records.

4. Performance Analysis

4.1 Observed Metrics

Total pages processed	**138**
Total processing time	~45 minutes
Average time per page	~26 seconds
Model load time (LightOnOCR)	~8 seconds
Prefill time per page	~3-5 seconds
Generation time per page	~15-20 seconds (up to 1024 tokens)

4.2 Optimisation Opportunities

- Batch inference: LightOnOCR supports batching processing 4 pages simultaneously could give 3-4x speedup.
- Split processing: A small classifier model is used to see if the page is handwriting or print. Then handwritten pages are sent to the cpu/igpu and the printed pages get processed on the gpu.

References:

[1] . https://huggingface.co/lightonai/LightOnOCR-2-1B-bbox-soup

[2] . https://huggingface.co/Riksarkivet/trocr-base-handwritten-swe/tree/main

[3] . https://huggingface.co/vikhyatk/moondream1